

Algebra: Chapter 0, Aluffi

Part 6: Linear Algebra

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§1 Free Modules Revisited

2.

$\text{tr}(aM) = a \text{tr}(M)$, so $\mathfrak{sl}_n(\mathbb{R})$ and $\mathfrak{sl}_n(\mathbb{C})$ are $\mathbb{R}$ -vector spaces by the natural action $a \cdot M := aM$. The following gives an $\mathbb{R}$ -basis for $\mathfrak{sl}_n(\mathbb{R})$:

$$\{e_{kl} \mid k \neq l\} \cup \{e_{11} - e_{kk} \mid k > 1\} \quad (1)$$

where e_{kl} is the $n \times n$ matrix with a 1 in the (k, l) th entry and zeros everywhere else. It is clear that the elements of (1) are linearly independent. It is also clear $\{e_{kl} \mid k \neq l\}$ span the nondiagonal entries of all matrices in $\mathfrak{sl}_n(\mathbb{R})$. The following shows that

$$\{e_{11} - e_{kk} \mid k > 1\} \quad (2)$$

span the diagonal entries of all the elements in $\mathfrak{sl}_n(\mathbb{R})$, which completes the verification that (1) spans $\mathfrak{sl}_n(\mathbb{R})$.

Induct on n : for $n = 2$, it is clear that $\begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$ spans all possible diagonal entries for the elements of $\mathfrak{sl}_2(\mathbb{R})$.

Suppose the proposition is true for $n - 1$. Take any $M = x_1 e_{11} + x_2 e_{22} + \dots + x_n e_{nn} \in \mathfrak{sl}_n(\mathbb{R})$; that is $x_1 + x_2 + \dots + x_n = 0$. Then the manipulation

$$M = \begin{bmatrix} x_1 + x_n & & & \\ & x_2 & & \\ & & \ddots & \\ & & & x_{n-1} \\ & & & & 0 \end{bmatrix} - x_n \begin{bmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \\ & & & & -1 \end{bmatrix}, \quad (3)$$

the natural injection of $\mathfrak{sl}_{n-1}(\mathbb{R})$ into the first $(n - 1)$ rows and columns of $\mathfrak{sl}_n(\mathbb{R})$, and the observation that the $(n - 1) \times (n - 1)$ submatrix of the first matrix on the right hand side of (3) is in $\mathfrak{sl}_{n-1}(\mathbb{R})$ shows that M is generated by (2).

Hence (1) is a basis for $\mathfrak{sl}_n(\mathbb{R})$; accordingly,

$$\dim \mathfrak{sl}_n(\mathbb{R}) = n^2 - n + (n - 1) = n^2 - 1.$$

The basis for $\mathfrak{sl}_n(\mathbb{C})$ is given similarly, except one has to remember $\mathbb{C}$ is a two-dimensional vector space over $\mathbb{R}$:

$$\{e_{kl} \mid k \neq l\} \cup \{ie_{kl} \mid k \neq l\} \cup \{e_{11} - e_{kk} \mid k > 1\} \cup \{ie_{11} - ie_{kk} \mid k > 1\} \text{ gives a basis,}$$

$$\dim \mathfrak{sl}_n(\mathbb{C}) = 2n^2 - 2$$

where $i := \sqrt{-1}$.

Next, $a(M + M^t) = aM + aM^t$ shows that $\mathfrak{so}_n(\mathbb{R})$ is also $\mathbb{R}$ -vector spaces by the natural action $a \cdot M := aM$. $M + M^t = 0 \iff a_{kl} = -a_{lk}$ for all k, l ; namely, $M + M^t = 0 \implies a_{kk} = 0$ for all k , so any real matrix that satisfies $M + M^t = 0$ is a member of $\mathfrak{sl}_n(\mathbb{R})$.

$$\{e_{kl} - e_{lk} \mid k < l\}$$

gives a basis for $\mathfrak{so}_n(\mathbb{R})$. Accordingly,

$$\dim \mathfrak{so}_n(\mathbb{R}) = \sum_{k=1}^n \sum_{l=k+1}^n 1 = \frac{n(n-1)}{2}$$

Proceed similarly with $\mathfrak{su}(n)$. The natural action $a \cdot M := aM$ makes $\mathfrak{su}(n)$ an $\mathbb{R}$ -vector space. $M + M^* = 0 \iff a_{kl} = -\overline{a_{lk}}$. Particularly, each a_{kk} only has a complex part, and one mustn't forget that $\mathfrak{su}(n)$ is a subset of $\mathfrak{sl}_n(\mathbb{C})$. Hence

$$\{e_{kl} - e_{lk} \mid k < l\} \cup \{ie_{kl} + ie_{lk} \mid k < l\} \cup \{ie_{11} - ie_{kk} \mid k > 1\}$$

gives a basis for $\mathfrak{su}(n)$ over $\mathbb{R}$, and

$$\dim \mathfrak{su}(n) = n(n-1) + (n-1) = (n+1)(n-1)$$

5.

(a)

This is false. As mentioned in the text, $\{2\}$ doesn't is a maximal linearly independent subset of $\mathbb{Z}$ as a free module over itself, but it doesn't generate $\mathbb{Z}$. Hence $\{2\}$ can never be completed into a basis for $\mathbb{Z}$; adding any other element will make the set linearly dependent.

(b)

By a similar argument as part (a), this is false. $\{2, 3\}$ generate $\mathbb{Z}$ as a module over itself because the two numbers are coprime. However, as above, neither $\{2\}$ nor $\{3\}$ are a basis for $\mathbb{Z}$; namely, they don't generate all of $\mathbb{Z}$.

7.

Assume $S \subset M$ is linearly independent over R . Suppose S is not linearly independent as a subset of V over K : there are $c_\alpha \in K$, finitely many of which aren't zero, and

$$\sum c_\alpha s_\alpha = 0$$

Since only finitely many c_α aren't zero, one can cancel denominators: there is some $a \in R$ such that $ac_\alpha \in R \setminus 0$ for all α such that $c_\alpha \neq 0$:

$$\sum ac_\alpha s_\alpha = 0, \quad (ac_\alpha \in R \setminus 0 \text{ for } \alpha \text{ such that } c_\alpha \neq 0)$$

This gives a nontrivial R -linear dependence relation of elements of S , contradicting the assumption that $S \subset M$ is linearly independent over R . Hence S must be a linearly independent subset of V over K .

The reverse direction is clear. Hence a maximal linearly independent subset of M is a maximal linearly independent subset of V . A maximal linearly independent subset of V is a basis for V , so if B is such a set,

$$\text{rank } M = \#B = \dim V$$

Suppose S generates M over R . That is, for all $\sum r_\alpha a_\alpha \in M$,

$$\sum r_\alpha a_\alpha = \sum r'_\beta s_\beta$$

for some $r'_\beta \in R$ with at most finitely many of them nonzero. Given any $\sum c_\alpha a_\alpha \in V$, one can cancel the denominators of the nonzero coefficients to get $\sum c_\alpha a_\alpha = \frac{1}{b} \sum bc_\alpha a_\alpha$ with $b \in R$ and $bc_\alpha \in R$. Since $\sum bc_\alpha a_\alpha$ can be treated as an element of $M \subset V$,

$$\sum c_\alpha a_\alpha = \frac{1}{b} \sum r_\gamma s_\gamma = \sum \frac{r_\gamma}{b} s_\gamma$$

for some $r_\gamma \in R$ with at most finitely many of them nonzero. Hence S generates V as a K -vector field.

However, $\langle S \rangle = V$ over K does not entail $\langle S \rangle = M$ over R . The case $M = \mathbb{Z}$ as a $\mathbb{Z}$ -module, $V = \mathbb{Q}$ as a $\mathbb{Q}$ -module, and $S = \{2\}$ serves as an easy counterexample.

8.

Suppose $\Phi : F^R(A) \xrightarrow{\sim} F^R(B)$ is an isomorphism. A is a basis for $F^R(A)$, so it is particularly a maximal linearly independent set. Since the two free modules are isomorphic, $\Phi(A)$ is then a maximal linearly independent set for $F^R(B)$, which means $\#A = \#B$, i.e., $A \cong B$ as sets.

The reverse direction holds by universal property arguments.